



PREDICTING OSTEOPOROSIS RISK

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OBJECTIVE

Develop a machine learning model to predict osteoporosis risk using demographic, lifestyle, and health related factors.

- Identify patterns and factors associated with osteoporosis risk.
- Compare multiple machine learning classification models.
- Develop and evaluate a tuned model for reliable risk prediction.

DATASET OVERVIEW

- Dataset size : 1958 records
- Original features : 16 columns
- Target variable : Osteoporosis
- Feature categories :
 - Demographic : Age, Gender, Race/Ethnicity
 - Lifestyle : Physical Activity, Smoking, Alcohol Consumption
 - Health-related : Family History, Calcium Intake, Vitamin D Intake, Prior Fractures
 - Medical : Medical Conditions, Medications, Hormonal Changes, Body Weight

DATA CLEANING & PREPROCESSING

1. Data Cleaning

- Checked dataset structure and data types.
- Identified missing values and replaced it with “Unknown” where appropriate.
- Checked for duplicates.
- Investigated repeated IDs.

2. Feature Preparation

- Removed ID because it is an identifier, not a predictive feature.
- Removed Age Group because it was created only for EDA.
- Separated features (X) and target (y) .

3. Encoding & Splitting

- Applied One-Hot Encoding to categorical variables.
- Split data into 80% training(1566 records) and 20% testing (392 records).
- Used stratified splitting to maintain class balance.

EXPLORATORY DATA ANALYSIS (EDA)

Age Distribution :

- Age ranged from 18 to 90 years.
- Mean age : 39.1 years.
- A large proportion of the dataset is in the 18-29 age group.

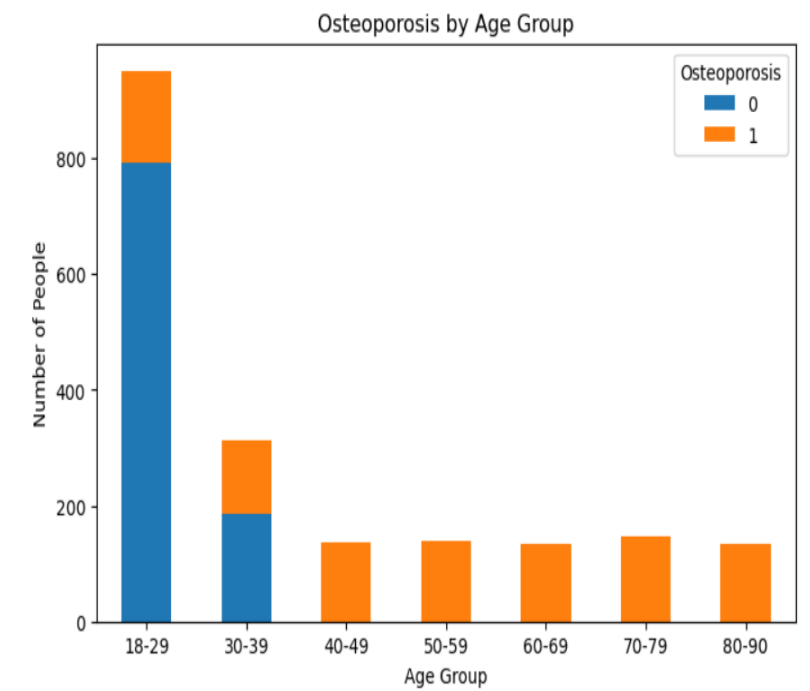
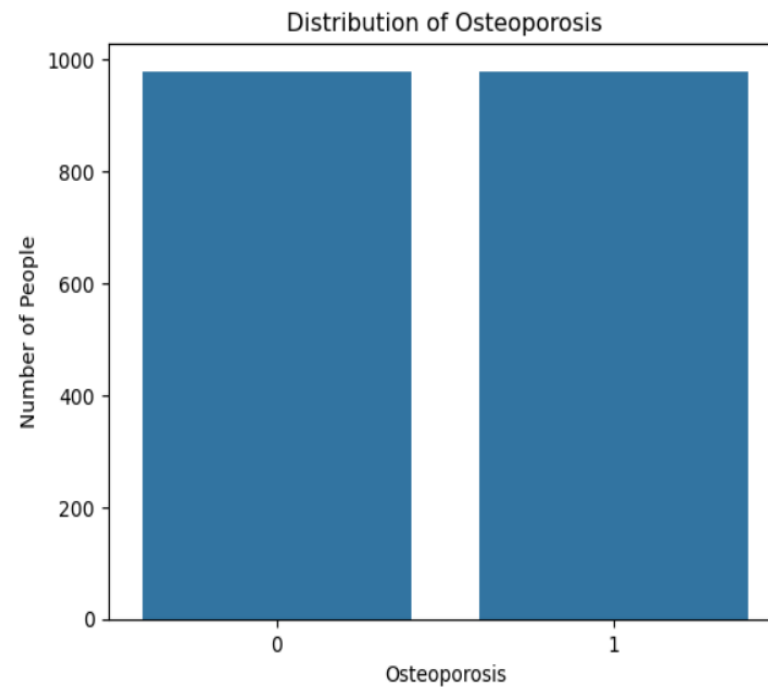
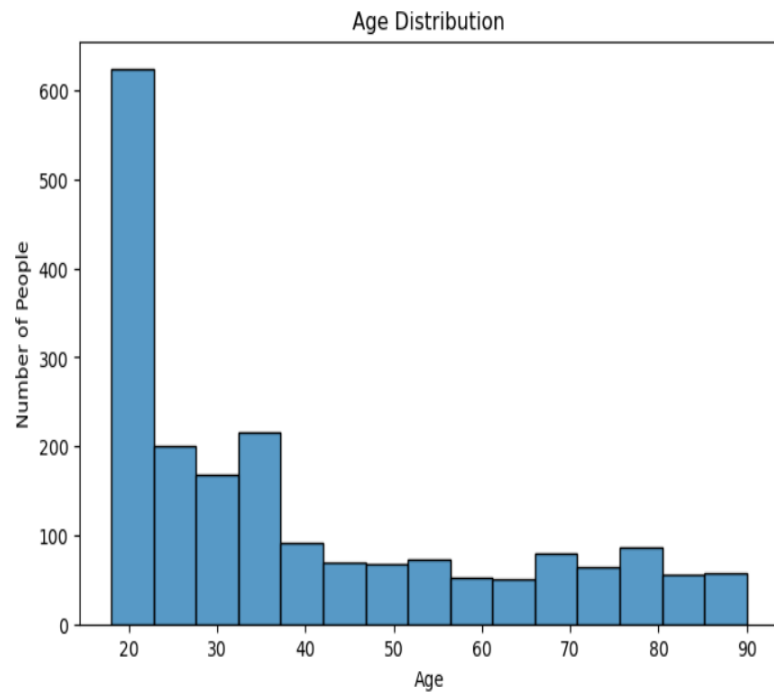
Target Distribution :

- 979 records without osteoporosis.
- 979 records with osteoporosis.
- Target classes were perfectly balanced.

Important dataset pattern :

A strong relationship was observed between age and the target variable. In this dataset , all records aged 40 years and above were labelled as having osteoporosis.

EXPLORATORY DATA ANALYSIS (EDA)



MACHINE LEARNING MODELS

Four **classification models** were trained and evaluated to predict osteoporosis risk.

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. K-Nearest Neighbors (KNN)

Evaluation Metrics :

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	80.9 %	85.8 %	74.0 %	79.5 %
Decision Tree	81.9 %	83.8 %	79.1 %	81.4 %
Random Forest	82.9 %	94.5 %	69.9 %	80.4 %
KNN	83.4 %	94.6 %	70.9 %	81.0 %

Key Observations:

- KNN achieved the highest initial test accuracy : 83.4 %
- Random Forest achieved the highest initial ROC-AUC : 88.6 %
- Decision Tree achieved the highest initial recall : 79.1 %

CROSS-VALIDATION

5-Fold Cross-validation

Model	CV Accuracy
Logistic Regression	82.70 %
Decision Tree	85.25 %
Random Forest	86.08 %
KNN	87.29 %

KNN HYPER PARAMETER TUNING

Method : GridSearchCV with 5-fold cross-validation

Parameters tested:

- Number of neighbors :3,5,7,9,11,15,20
- Weights : Uniform, Distance
- Distance Metric : Euclidean, Manhattan

Best Parameters:

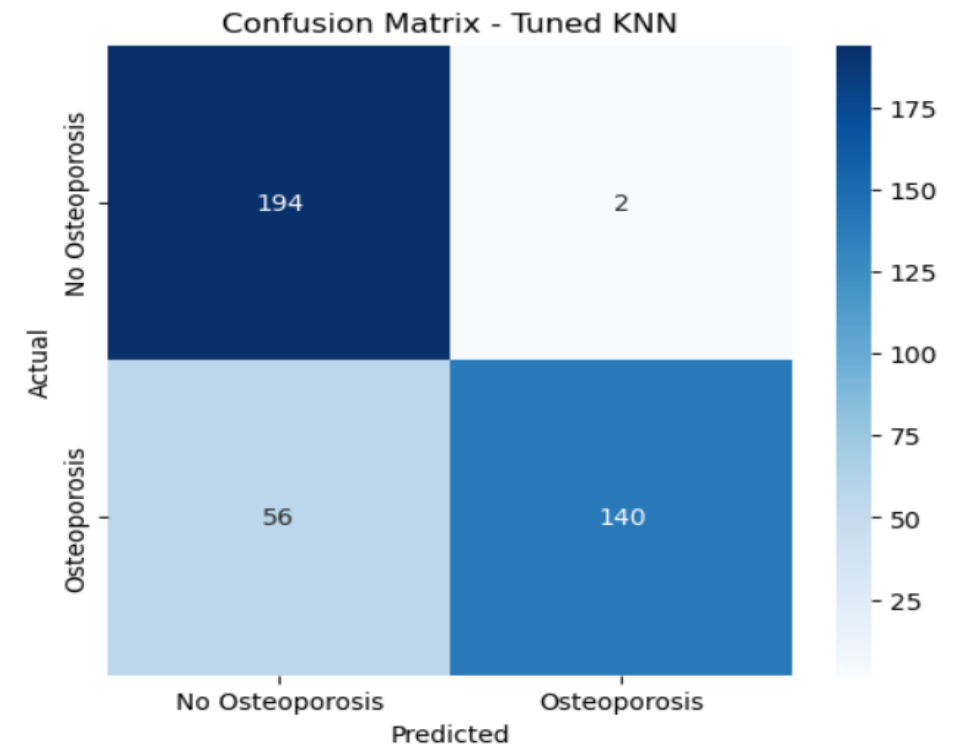
- Neighbors : 15
- Weights : Uniform
- Metric : Euclidean
- Tuned CV Accuracy : 88.44 %

FINAL MODEL - TUNED KNN

Selected Model : K-Nearest Neighbors (KNN)

Performance:

Metric	Result
Test Accuracy	85.20 %
Precision	99 %
Recall	71 %
F1-Score	83 %
ROC-AUC	87.63 %
5-Fold CV Accuracy	88.44 %



KEY FINDINGS

- Age was the most influential feature in the Random Forest model.
 - Age accounted for approximately 64.6 % of the model's feature importance.
 - Other individual features had much smaller important values.
 - This finding is consistent with the strong age-target relationship observed during EDA.
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- ❖ Feature importance indicates how the Random Forest model used the features. It does not prove that a feature causes osteoporosis.

LIMITATIONS & POTENTIAL APPLICATIONS

Limitations

- Strong age-target relationship : All records aged 40 + were labelled as having osteoporosis.
- Dataset size : Only 1958 records were available.
- Generalizability : Results may not represent other populations or datasets.
- Recall : The final KNN model had 71 % recall , meaning some actual osteoporosis cases were missed.
- Clinical validation : The model has not been clinically validated.

Potential Healthcare Application

- Support early risk identification.
- Help identify individuals who may need further assessment.
- Support preventive follow-up decisions.
- Could serve as a decision-support tool after proper validation

CONCLUSION

Project Summary

- Developed an end-to-end machine learning classification model for osteoporosis risk prediction.
- Performed data cleaning, EDA, preprocessing, encoding and model evaluation.
- Compared four models : Logistic Regression, Decision Tree, Random Forest and KNN.
- Used 5-fold cross-validation and GridSearchCV to improve model evaluation and tune KNN.
- Selected Tuned KNN as the final model.

Final model Performance:

- 85.2% Test Accuracy
- 99% Precision
- 71% Recall
- 83% F1-Score
- 87.6% ROC-AUC
- 88.4% 5-Fold CV Accuracy



THANK YOU